

# 15-normal-subgroup: 正规子群 (Normal Subgroup)

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2026 年 06 月 18 日



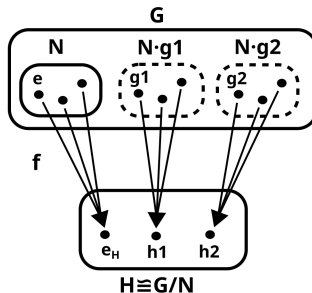
# Fundamental Homomorphism Theorems



Emmy Noether (1882 ~ 1935)

# Fundamental Homomorphism Theorems

## Homomorphism theorem



● element of a group

□ group    □ not a group

$N$  normal subgroup

$e$  identity of  $G$      $e_H$  identity of  $H$

$f$  homomorphism  $G \rightarrow H$

Help us understand the structure of a group  
via its normal subgroups

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$H = \{(1), (1\ 2)\} \leq S_3$$

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It is possible that  $aH \neq Ha$ .

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$$\forall a \in S_3 \ aH = Ha$$

## Definition (Normal Subgroup (正规子群))

Suppose that  $H \leq G$ . If

$$\forall a \in G \ aH = Ha,$$

then  $H$  is a **normal subgroup** of  $G$ , denoted  $H \triangleleft G$ .

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$$aH = Ha \not\Rightarrow \forall h \in H \ ah = ha$$

$$aH = Ha \Rightarrow \forall h \in H \ \exists h' \in H \ ah = h'a$$

## Definition (正规子群的陪集)

Suppose that  $H \triangleleft G$ .

$$G/H = \{aH \mid a \in G\}$$

is the **set of cosets** of  $H$  in  $G$ .

## Definition (Quotient Group (商群))

Suppose that  $H \triangleleft G$ . Define

$$aH \cdot bH = (ab)H.$$

Then  $(G/H, \cdot)$  is a **group**, called the **quotient group** of  $G$  by  $H$ .

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$$\begin{aligned} a'H \cdot b'H &= (a'b')H \\ &= a'(b'H) = a'(bH) = a'(Hb) \\ &= (a'H)b = (aH)b = a(Hb) = a(bH) \\ &= (ab)H = aH \cdot bH \end{aligned}$$

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$$G/H = \{(1)H, (1\ 2)H\} \cong \mathbb{Z}_2$$

$$G = (\mathbb{Z}, +) \quad H = (6\mathbb{Z}, +) \triangleleft G$$

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## Theorem

Suppose that  $H \leq G$ .

$$H \triangleleft G \iff \forall a \in G \ aHa^{-1} = H \iff \boxed{\forall a \in G, h \in H \ aha^{-1} \in H}$$

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$$\boxed{\forall a \in G \ (aH = Ha \rightarrow \forall h \in H \ aha^{-1} \in H)}$$

$$\begin{aligned} aH = Ha &\implies aHa^{-1} = (Ha)a^{-1} = H(aa^{-1}) = H \\ &\implies aHa^{-1} \subseteq H \\ &\implies \forall h \in H \ aha^{-1} \in H \end{aligned}$$

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$$\boxed{\forall a \in G, h \in H \ (aha^{-1} \in H \rightarrow aH = Ha)}$$

$$\begin{aligned} aha^{-1} \in H &\implies ah = (aha^{-1})a \in Ha \implies aH \subseteq Ha \\ a^{-1}ha = a^{-1}h(a^{-1})^{-1} \in H &\implies ha \in aH \implies Ha \subseteq aH \end{aligned}$$

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$$\forall \sigma \in S_3, \tau \in H \quad \sigma\tau\sigma^{-1} \in H$$

## Theorem

$$\sigma\tau\sigma^{-1} = \begin{pmatrix} \sigma(1) & \sigma(2) & \dots & \sigma(n) \\ \sigma(\tau(1)) & \sigma(\tau(2)) & \dots & \sigma(\tau(n)) \end{pmatrix}$$

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$$(1\ 2)(1\ 2\ 3)(1\ 2)^{-1} = \begin{pmatrix} 2 & 1 & 3 \\ 1 & 3 & 2 \end{pmatrix} = (1\ 3\ 2)$$

## Definition (Homomorphism (同态))

Let  $(G, \cdot)$  and  $(G', *)$  be two groups. Let  $\phi$  be a function such that

$$\forall a, b \in G \quad \phi(a \cdot b) = \phi(a) * \phi(b).$$

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Then  $\phi$  is a **homomorphism** from  $G$  to  $G'$ .

If  $\phi$  is a bijection, then  $G$  and  $G'$  are called **isomorphic**.

$$\phi : G \cong G'$$

$$\begin{aligned}\phi : (\mathbb{Z}, +) &\rightarrow (\mathbb{R}^*, *) \\ n &\mapsto (-1)^n\end{aligned}$$



$$\phi : (\mathbb{Z}, +) \rightarrow (\mathbb{R}^*, *)$$

$$n \mapsto (-1)^n$$

$$\phi(m + n) = (-1)^{m+n} = \phi(m)\phi(n)$$

$$\begin{aligned}\phi : (\mathbb{Z}, +) &\rightarrow (\mathbb{Z}_6, +_6) \\ a &\mapsto [a]_6\end{aligned}$$

$$\phi(a + b) = [a + b]_6 = [a]_6 +_6 [b]_6 = \phi(a) +_6 \phi(b)$$

$$\begin{aligned}\phi : \mathbb{R}[x] &\rightarrow \mathbb{R}[x] \\ f(x) &\mapsto f'(x)\end{aligned}$$

$\mathbb{R}[x]$  : 全体实系数多项式关于多项式的加法构成的群

$$\phi(f(x) + g(x)) = (f(x) + g(x))' = f'(x) + g'(x) = \phi(f(x)) + \phi(g(x))$$

## Theorem

*Suppose that  $\phi$  is a homomorphism from  $G$  to  $G'$ .*

*Let  $e$  and  $e'$  be identities of  $G$  and  $G'$ , respectively.*

$$(1) \quad \phi(e) = e'$$

$$(2) \quad \phi(a^{-1}) = (\phi(a))^{-1}$$

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$$\phi(a)\phi(a^{-1}) = \phi(aa^{-1}) = \phi(e) = e' = \phi(a)(\phi(a))^{-1}$$

## Theorem

Suppose that  $\phi$  is a homomorphism from  $G$  to  $G'$ .

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$$H \leq G \implies \phi(H) \leq \phi(G)$$

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(3)

$$K \leq G' \implies \phi^{-1}(K) \leq G$$

(4)

$$K \triangleleft G' \implies \phi^{-1}(K) \triangleleft G$$



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Definition (同态核 (Kernel))

Suppose that  $\phi$  is a homomorphism from  $G$  to  $G'$ .

Let  $e'$  be the identity of  $G'$ . Then  $\{e'\} \triangleleft G'$ .

$$\phi^{-1}(\{e'\}) = \{a \in G \mid \phi(a) = e'\} \triangleleft G$$

is the **kernel** of  $\phi$ , denoted  $\text{Ker } \phi$ .

$$\phi : \mathbb{Z} \rightarrow \mathbb{R}^*$$

$$n \mapsto (-1)^n$$

$$\text{Ker } \phi =$$

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$$\text{Ker } \phi = 2\mathbb{Z}$$



$$\phi : \mathbb{Z} \rightarrow \mathbb{Z}_6$$

$$a \mapsto [a]_6$$

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$$\begin{aligned}\phi : \mathbb{R}[x] &\rightarrow \mathbb{R}[x] \\ f(x) &\mapsto f'(x)\end{aligned}$$

$\mathbb{R}[x]$  : 全体实系数多项式关于多项式的加法构成的群

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$$\text{Ker } \phi = \mathbb{R}$$

## Theorem (Fundamental Homomorphism Theorem (群同态基本定理))

Suppose that  $\phi$  is a homomorphism from  $G$  to  $G'$ . Then

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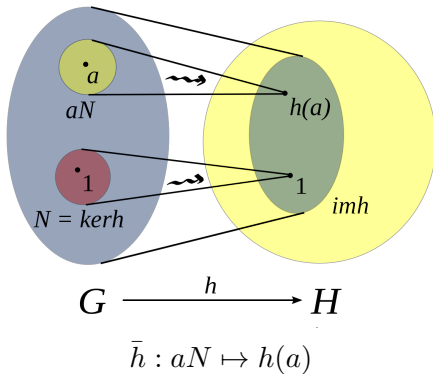
同态核可以看作群  $G$  与其同态像  $\phi(G)$  之间**相似程度**的一种度量

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同态核可以看作群  $G$  与其同态像  $\phi(G)$  之间**相似程度**的一种度量



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$$\mathbb{Z}/(2\mathbb{Z}) = (2\mathbb{Z}, 2\mathbb{Z} + 1) \cong \phi(\mathbb{Z}) = \{-1, 1\} \cong \mathbb{Z}_2$$

$$\phi : \mathbb{Z} \rightarrow \mathbb{Z}_6$$

$$a \mapsto [a]_6$$

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$$\text{Ker } \phi = \mathbb{R}$$

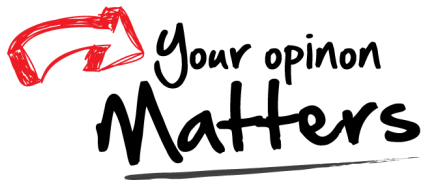
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$$\text{Ker } \phi = \mathbb{R}$$

$$\mathbb{R}[x]/\mathbb{R} \cong \phi(\mathbb{R}[x]) = \mathbb{R}[x]$$

Thank  
You!



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